INCH-POUND

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10 August 1984

MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14A DUAL OUTPUT AMPLIFIER MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the Dual Output Amplifier Module described herein shall consist of this specification and the latest issue of MIL-I-85846.

REQUIREMENTS:

Item description: The Dual Output Amplifier Module shall provide two amplifiers having a high degree of electrical isolation between the two inputs and between the two outputs. The amplifier shall provide amplification of audio signals with gain characteristics as specified herein when used in Audio Frequency Amplifier AM-3889/AIC-14A.

Item diagram: The Dual Output Amplifier shall provide the functions represented schematically in accordance with the diagram of Figure 1.

Interface description: The Dual Output Amplifier input and output signals will normally display the characteristics presented in Table I.

Environmental: The Dual Output Amplifier shall meet the environmental requirements of MIL-I-85846 except the fluids requirement shall not apply.

Characteristics:

Performance characteristics:

Input characteristics:

Audio No. 1 (AUDIO NO. 1): The Dual Output Amplifier shall have an AUDIO NO. 1 input circuit having the following performance characteristics:

Frequency and Amplitude: The amplifier shall accept AUDIO NO. 1 input drive signals which range from 300 to 7000 Hz and from 1.2 to 15.0 Vac in amplitude. The amplifier shall be capable of withstanding overvoltage inputs of up to 20.0 Vac without sustaining damage.

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TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
Audio No. 1 (AUDIO NO. 1)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 15.0 Vac. No dc component normally present.
Audio No. 2 (AUDIO NO. 2)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 15.0 Vac. No dc component normally present.
Power (PWR)	+27.5 ± 0.5 Vdc.
Outputs:	
600 Ohm No. 1 (600 OHM NO. 1)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 35 Vac. Signal power range is 0 to 2 W.
600 Ohm No. 2 (600 OHM NO. 2)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 35 Vac. Signal power range is 0 to 2 W.
7.5 Ohm No. 1 (7.5 OHM NO. 1)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 4.0 Vac. Signal power range is 0 to 2 W.
7.5 Ohm No. 2 (7.5 OHM NO. 2)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 4.0 Vac. Signal power range is 0 to 2 W.
Power (PWR GND)	Power return. See MIL-I-85846.
System (SYS GND)	Signal return. See MIL-I-85846.

Input impedance: The amplifier shall have an input impedance of 2400 ohms ± 10 percent.

Isolation: The isolation between AUDIO NO. 1 and AUDIO NO. 2 signal inputs shall be not less than 20 dB.

Audio No. 2 (AUDIO NO. 2): The AUDIO NO. 2 input characteristics shall be as specified for AUDIO NO. 1.

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Power (PWR): The amplifier shall accept and operate within the constraints of this specification when supplied with electrical power as specified in MIL-I-85846. The maximum load presented by the amplifier shall be not greater than 350 milliamperes.

Output characteristics: The amplifier shall have the following output characteristics.

600 ohm No. 1 (600 OHM NO. 1): The 600 OHM NO. 1 output shall have the following characteristics:

Frequency and Amplitude: The output shall be within ±1 dB of the input amplitude at 1000 Hz and in accordance with the response curve of Figure 2. Within the frequency range from 300 to 7000 Hz, the signal amplitude shall be flat within ±1 dB of the reference value at 1000 Hz when terminated in a 600 ohm resistive load. The output shall be 8 + 2 dB (.12-.19 volts) with a nominal input of 0.06 V at 1000 Hz applied to the AUDIO NO. 1 input. The input/output characteristics (gain) shall be linear within ±1 dB for input levels between 0 and 15 volts rms.

Distortion: When operating at full rated power output, the total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than five percent at standard conditions as specified in MIL-I-85846. The total harmonic distortion over the frequency range from 300 to 7000 Hz shall be not greater than 10 percent over the entire environmental envelope as specified in MIL-I-85846.

Output power: The output of the amplifier shall be capable of delivering 2 watts into a 600 ohm or 7.5 ohm resistive load.

Output stability: The output shall show no evidence of regeneration or oscillation throughout the environmental envelope as specified in MIL-I-85846 under any combination of input-output open, short or complex impedance terminations. The Dual Output Amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

Internal noise: The Dual Output Amplifier shall have a 600 OHM NO. 1 output amplitude of less than 5 millivolts when operated under standard conditions as specified in MIL-I-85846 with the input circuit shorted to system ground.

Susceptibility to Stray Magnetic Fields: The amplifier shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2800 Hz, the output from the amplifier shall be not greater than 5 millivolts at 400 Hz. Between 400 and 2800 Hz the output shall be not greater than a linear increase up to 50 millivolts at 2800 Hz.

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Isolation: When the input circuit is properly terminated but no signal is applied, any signal appearing on its terminated output line shall be considered as undesired signal. The undesired signal level on this output line shall be not less than 60 dB below its rated voltage output for a normal signal. Undesired signal level on this output line shall be measured with the amplifier properly terminated and the other channel driven to rated voltage output.

Phasing: The phase difference between the input and output shall be 180 ± 20 degrees.

600 ohm No. 2 (600 OHM NO. 2); The 600 OHM NO. 2 output shall have characteristics identical to those described for the 600 OHM NO. 1 output.

7.5 ohm No. 1 (7.5 OHM NO. 1): The 7.5 OHM NO. 1 output shall have the following characteristics:

Frequency and Amplitude: Within the frequency range from 300 to 7000 Hz the signal amplitude shall be flat within ±1 dB when terminated in a 7.5 ohm resistive load. The output shall be below a nominal input of 13.4 Vac at 1000 Hz applied to AUDIO input by 11 ± 2 dB (3.0 to 4.8 Vac) when terminated in a 7.5 ohm load. The input/output characteristics (gain) shall be linear within ±1 dB for input levels between 0 and 15 volts rms.

Distortion: The total harmonic distortion over the frequency range of 300 to 7000 Hz shall be not greater than the values listed in Table II below.

TABLE II. Maximum harmonic distortion.

Condition	Output Level (Watts)	Distortion Limit (Percent)
Standard	2	5
All Environmental	2	10

Where:
Standard - Standard conditions as specified in MIL-I-85846.
All - The entire environmental envelope specified in MIL-E-5400 for Class 2 equipment as modified by MIL-I-85846.

Output stability: The output shall show no evidence of regeneration or oscillation throughout the environmental envelope of MIL-I-85846 under any combination of input-output open, short or complex impedance terminations. The Dual Output Amplifier shall be capable of withstanding any combination of open/short circuit terminations without damage and resume operation within specification when properly terminated.

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Internal noise: The amplifier shall have an output of less than 5 millivolts when operated under standard conditions as specified in MIL-I-85846 with the input circuit shorted to system ground.

Susceptibility to Stray Magnetic Fields: The Dual Output Amplifier shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2800 Hz, the output from the Dual Output Amplifier shall be not greater than 5 millivolts at 400 Hz. Between 400 and 2800 Hz the output shall be not greater than a linear increase up to 50 millivolts at 2800 Hz.

Isolation: When the input circuit is properly terminated but no signal is applied, any signal appearing on this outgoing line shall be considered as undesired signal. The undesired signal level on this outgoing line shall be not less than 60 dB below rated power output for a normal signal. Undesired signal level on this outgoing line shall be measured with the amplifier properly terminated and the other channel driven to rated power output.

Phasing: The phase difference between the input and output shall be 180 $\pm$ 20 degrees.

7.5 ohm No. 2 (7.5 OHM NO. 2): The 7.5 OHM NO. 2 output shall have characteristics identical to those described for the 7.5 OHM NO. 1 output.

Physical characteristics:

Dimensions: The amplifier shall consist of a single encapsulated module having the dimensions of Figure 3. All outside dimensions shall be within $\pm$0.01 inch of the nominal value shown.

Weight: Weight of the amplifier shall be not greater than 14 ounces.

Design and construction: The design and construction of the amplifier shall be in accordance with the requirements of 3.3 of MIL-I-85846(AS).

QUALITY ASSURANCE PROVISIONS: See MIL-I-85846.

Responsibility for tests and inspections: See MIL-I-85846.

Test methods:

Examination of product: See MIL-I-85846. Units procured as spares shall not be subjected to the Fluids inspection. (See 4.2.1.10 of MIL-I-85846.)

Operational test: See MIL-I-85846.

Input impedance: Verify that the AUDIO NO. 1 and AUDIO NO. 2 input impedances are in accordance with requirements herein.

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Gain: Verify that the amplifier gains at 1000 Hz is in accordance with requirements herein.

Frequency Response: Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements herein.

Linearity: Using a 1000 Hz input signal which is varied between 0.01 Vac and 15 Vac, verify that the amplifier meets the requirements herein.

Output power: With the amplifier properly terminated, apply a 1000 Hz signal of appropriate amplitude to verify the requirements herein.

Distortion: Verify the requirements herein.

Amplifier stability: Verify the requirements herein.

Internal noise: Verify the requirements herein.

Susceptibility to Stray Magnetic Fields: Verify the performance requirements herein.

Phasing: Verify the requirements herein.

Environmental stress screening (burn-in): An environmental stress screening inspection shall be required on all Amplifier Modules. Only those modules that have successfully completed the environmental stress screening inspection shall be delivered to the procuring activity. Prior to and after each submission to the screening inspection, a room ambient full electrical performance inspection shall be performed on each Amplifier Module. Each Amplifier Module shall be subjected to 10 4-hour temperature cycles consisting of a 2-hour cold period at -54°C and a 2-hour hot period at +71°C. The modules need not be energized during the temperature cycling portion of this inspection. The thermal stress screening inspection chamber's temperature change rate shall be greater than 10°C per minute. The amplifier module shall be allowed a ± 3 dB gain (see Frequency and Amplitude paragraph herein) when subjected to hot/cold temperature cycling only.

Special tests: See MIL-I-85846.

Preparing Activity:
Navy - AS
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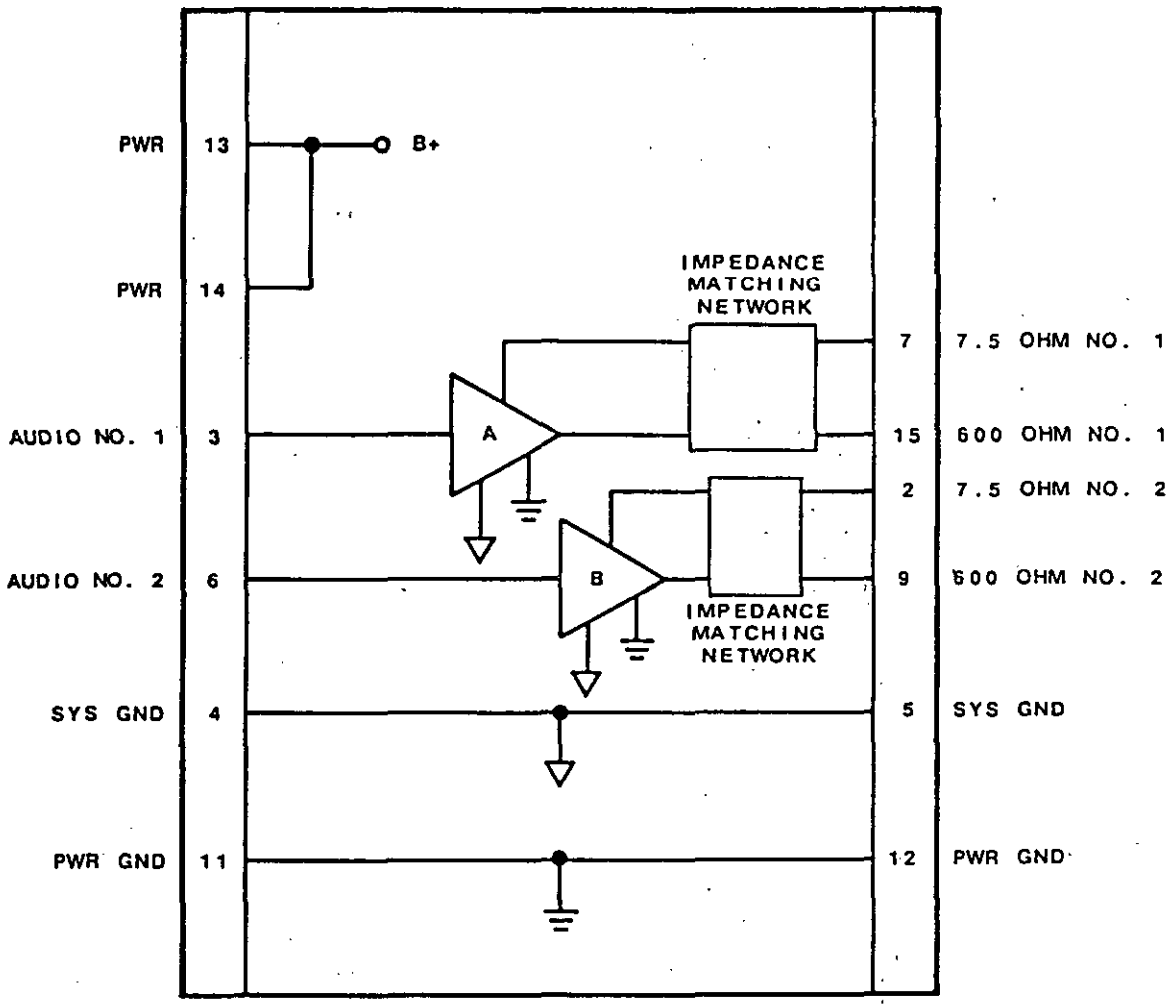
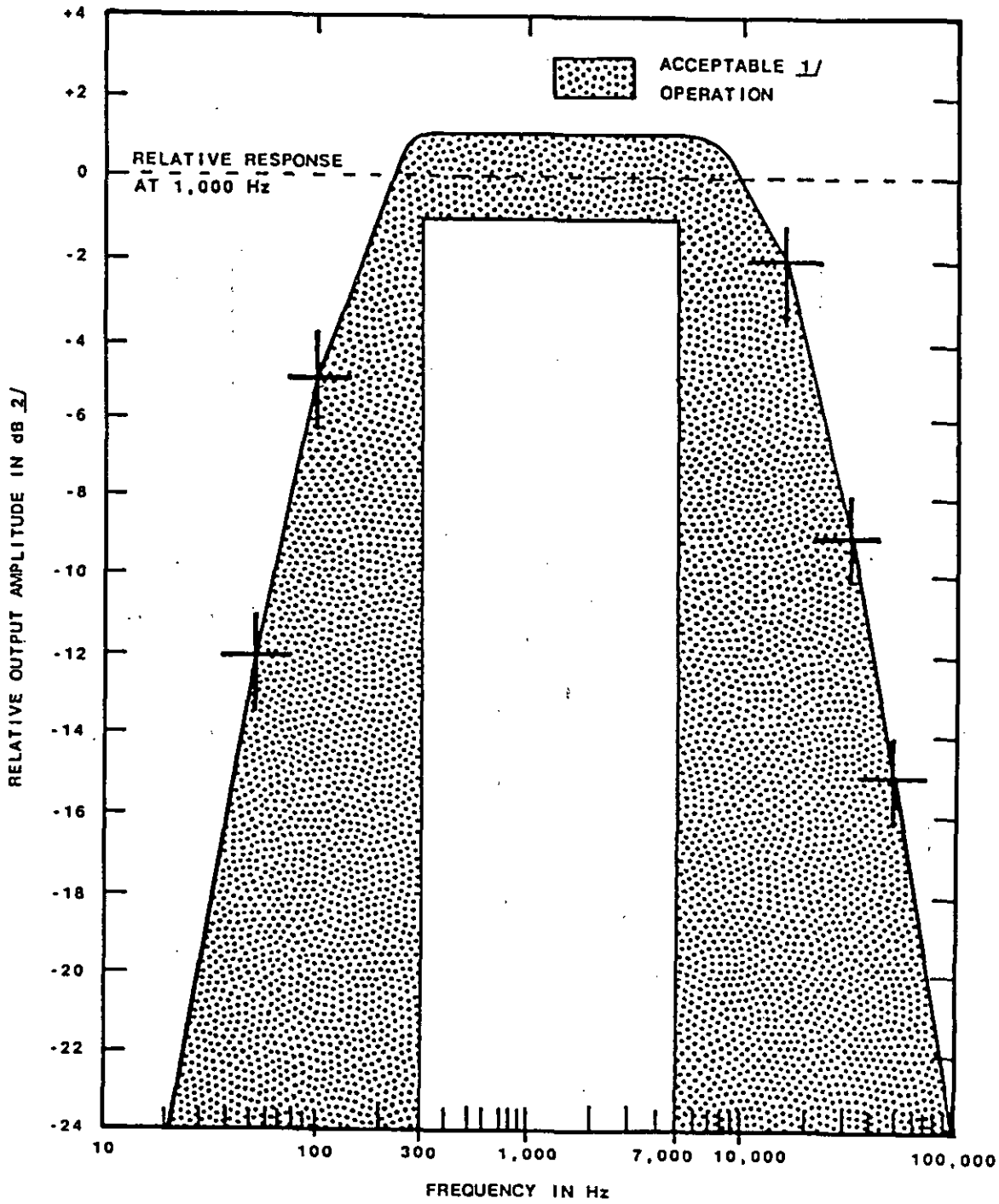


FIGURE 1. Dual output amplifier functional diagram.

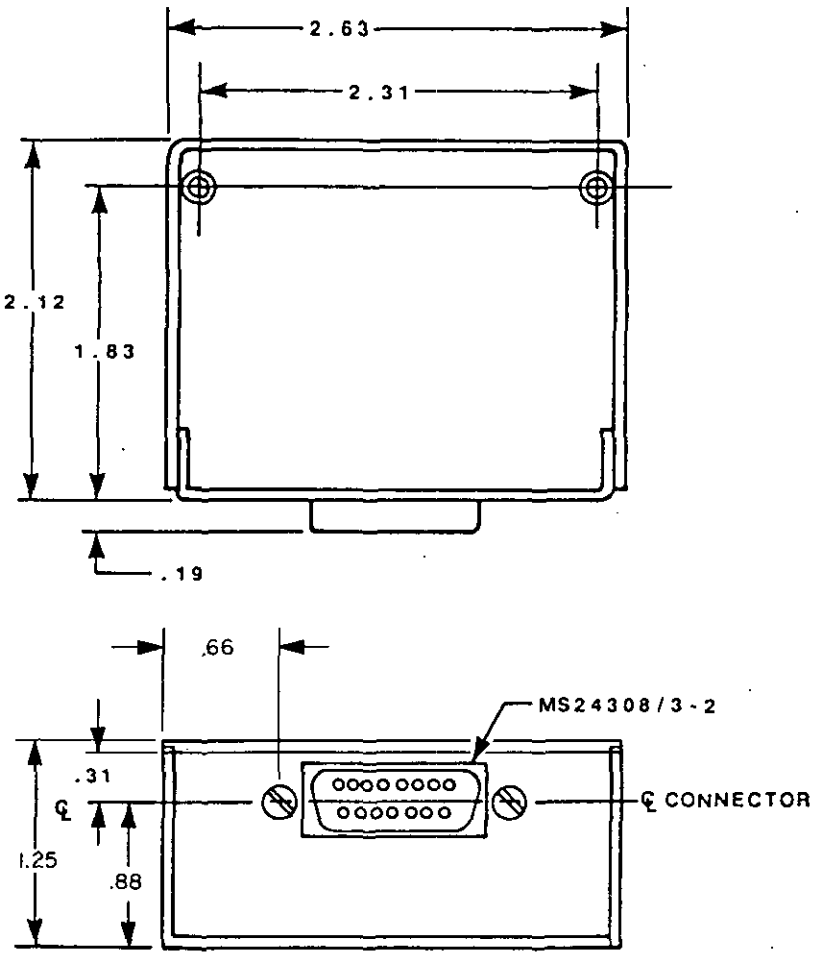
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1/ Termination into 600 ohms or 7.5 ohms.
2/ Output amplitude is measured with respect to the amplitude at 1000 Hz.

FIGURE 2. Audio frequency response.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.01 inches unless otherwise specified.

FIGURE 3. Dual output amplifier module outline drawing.

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